

CodeNova Business Website

Frontend Requirements & UI Specification

Version: 1.0 **Platform:** Responsive Web **Primary Goal:** High-conversion agency website

Purpose: This document defines the frontend pages, components, layout, visual system, interactions, responsive behavior, accessibility, SEO and frontend implementation standards for the CodeNova website.

1. Frontend Goals

- Present CodeNova as a professional software, AI and digital product development agency.
- Convert visitors into project inquiries, consultation bookings and quote requests.
- Provide a premium experience across mobile, tablet and desktop.
- Make services, portfolio, process and contact options easy to discover.
- Create SEO-friendly service and case-study pages that can also support advertising landing pages.

2. Recommended Frontend Stack

- Next.js with TypeScript.
- Tailwind CSS for styling and responsive layout.
- React components for reusable UI.
- Lucide React or another consistent icon library.
- Next/Image or equivalent image optimization.
- Framer Motion or lightweight CSS animations only where animation improves UX.
- React Hook Form plus Zod or equivalent for form handling and validation.

3. Brand & Visual Direction

- Style: modern, premium, minimal, technology-focused, trustworthy and business-oriented.
- Primary: #0F172A.
- Secondary: #2563EB.
- Accent: #06B6D4.
- Background: #F8FAFC.
- White: #FFFFFF.
- Text: #111827.
- Muted: #64748B.
- Use gradients sparingly and keep the visual hierarchy clean.
- Avoid excessive glassmorphism, excessive animation, fake statistics and crowded layouts.

4. Typography

- Use a modern sans-serif family such as Inter, Geist or another highly readable web font.
- Hero heading: large, bold and responsive.

- Section heading: strong hierarchy with short supporting copy.
- Body text: comfortable line height and readable contrast.
- Buttons: medium/bold weight with clear action wording.
- Use consistent typography tokens rather than page-specific arbitrary sizes.

5. Global Layout

- Sticky responsive navigation.
- Centered content container with consistent maximum width.
- Consistent horizontal page padding.
- Reusable section spacing tokens.
- Footer on every public page.
- Floating WhatsApp CTA on supported pages.
- Primary CTA should remain visually prominent without becoming intrusive.

6. Header / Navigation

- Desktop: CodeNova logo, Home, Services, Solutions, Portfolio, About, Process, Blog and a primary consultation CTA.
- Services and Solutions should use dropdown/mega-menu behavior where useful.
- Mobile: hamburger menu with expandable sections and prominent CTA.
- Header should remain accessible while scrolling.
- Active navigation item should be visually identifiable.

7. Homepage — Section Order

- Announcement/trust strip if needed.
- Hero.
- Trusted technology/service statement.
- Services overview.
- Why CodeNova.
- Featured portfolio.
- AI solutions.
- Business automation workflow.
- How we work.
- Industries.
- Technology stack.
- Testimonials only when genuine.
- Project estimator CTA.
- FAQ.
- Final consultation CTA.
- Footer.

8. Hero Section

- Headline: "We Build Digital Products That Move Businesses Forward."
- Supporting message explaining websites, apps, AI agents, automation and custom software.
- Primary CTA: Start Your Project.
- Secondary CTA: Explore Our Work.
- Optional visual: modern product/dashboard/AI interface composition.
- Add concise trust indicators such as Custom Development, AI Solutions, Responsive Design and Business-Focused.

9. Services UI

- Use reusable service cards with icon, title, short description and CTA.
- Core services: Website Development, Web Applications, Mobile App Development, Desktop Application Development, AI Agents, AI Automation, E-Commerce, Custom Software and UI/UX Design.
- Cards should have consistent height and interaction behavior.
- Service detail pages should share a common layout template.

10. Desktop Application Development UI

- Dedicated route: /services/desktop-application-development.
- Hero should clearly state that CodeNova builds desktop applications for business and operational workflows.
- Show platform options such as Windows, macOS and cross-platform, subject to project requirements.
- Use-case cards: POS, billing, inventory, ERP, school systems, offline business tools, internal dashboards and custom utilities.
- Feature section: offline mode, local data, synchronization, printing, device integration, APIs and secure updates where applicable.
- CTA: Build My Desktop Application / Discuss My Project.

11. Service Detail Page Template

- Hero + service title.
- Problem statement.
- How CodeNova solves it.
- Key features.
- Business benefits.
- Technology/options.
- Development process.
- Related projects.
- FAQ.
- Project estimator CTA.
- Consultation CTA.

12. Solutions Section

- Business Automation.
- AI Customer Support.
- AI Lead Generation.
- CRM Solutions.
- Custom Business Systems.
- Present each solution as a business outcome rather than only a technology feature.

13. AI Solutions UI

- AI Agents.
- AI Chatbots.
- AI Voice Agents.
- AI Sales Agents.
- Lead Qualification.
- AI Document Processing.
- AI Business Assistants.
- AI Workflow Automation.
- Use diagrams or workflow animations to show how AI connects to business processes.

14. Portfolio UI

- Portfolio grid with category filters.
- Project card: image, name, category, short description, technology tags and View Case Study.
- Use genuine project information and screenshots.
- Support featured projects on homepage.
- Case-study pages should include a clear project narrative and final CTA.

15. Case Study UI

- Project hero.
- Overview.
- Problem.
- Solution.
- Feature highlights.
- Technology.
- UI screenshots.
- Development process.
- Results/business impact when verified.
- Related projects.
- CTA: Discuss a Similar Project.

16. Process UI

- 01 Discovery.
- 02 Strategy.
- 03 Design.
- 04 Development.
- 05 Testing.
- 06 Launch.
- 07 Support.
- Use a timeline or connected step layout that works on both desktop and mobile.

17. Industries UI

- Healthcare, Education, Retail, Restaurants, Finance, Real Estate, Entertainment, Logistics, Startups and Professional Services.
- Use compact industry cards linking to relevant solutions or landing pages.

18. Pricing UI

- Display starting prices only when CodeNova has verified pricing.
- Cards may include Website, Web Application, Mobile App, Desktop App, AI Solution and Custom Software.
- Use "Custom Quote" for solutions that depend heavily on scope.
- Add a clear disclaimer that final pricing depends on requirements.
- CTA: Get a Project Estimate.

19. Project Estimator UI

- Step 1: Select project type.
- Step 2: Select complexity.
- Step 3: Select required features.
- Step 4: Optional contact details.
- Step 5: Display indicative estimate.
- Features may include Authentication, Payments, Admin Panel, AI, Maps, Notifications, Chat, Analytics, API Integration and Database.
- Include progress indicator, back/next controls and clear validation.

20. Contact Page UI

- Headline: Let's Build Something Great.
- Fields: Name, Email, Phone, Company, Country, Service, Budget, Project Description and Timeline.
- Show validation messages next to invalid fields.
- On success, show confirmation state without requiring a full-page refresh.
- Provide alternative contact options: WhatsApp, email and phone when configured.

21. Consultation Booking UI

- Fields: Name, Email, Company, Project Type, Preferred Date, Preferred Time and Message.
- Calendar integration can be added later.
- Show available times only when a scheduling provider is connected.
- Show confirmation after successful booking.

22. Blog Frontend

- Blog listing with categories and search/filter.
- Blog cards with image, title, excerpt, category and date.
- Article page with readable typography, table of contents where useful, related posts and CTA.
- Support SEO title, meta description, Open Graph image and canonical URL.

23. Footer

- CodeNova logo and short company description.
- Services links.
- Solutions links.
- Portfolio, About, Process and Blog links.
- Contact details.
- Social links.
- Privacy Policy, Terms, Cookie Policy and Refund Policy.
- Copyright line.

24. Reusable Component Library

- Navbar.
- MobileMenu.
- Footer.
- Button.
- Badge.
- SectionHeader.
- ServiceCard.
- SolutionCard.
- PortfolioCard.
- CaseStudySection.
- TechnologyBadge.
- TestimonialCard.
- FAQAccordion.
- CTASection.
- ContactForm.

- ConsultationForm.
- EstimatorWizard.
- Modal.
- Toast/Notification.
- LoadingSkeleton.
- EmptyState.

25. Responsive Design

- Mobile-first implementation.
- Target widths: approximately 360px, 390px, 768px, 1024px, 1280px, 1440px and 1920px.
- Desktop grids should collapse cleanly into tablet and mobile stacks.
- Navigation converts to a mobile menu.
- Buttons should remain easy to tap on touch devices.
- Avoid horizontal overflow.
- Images, typography and spacing must scale appropriately.

26. Animations & Interactions

- Use subtle entrance animations for major sections.
- Use hover/focus states on cards and buttons.
- Use smooth scrolling for internal anchors where appropriate.
- Use micro-interactions for form submission and navigation.
- Respect prefers-reduced-motion.
- Avoid animations that delay access to content or reduce performance.

27. Accessibility

- Use semantic HTML.
- All form controls require accessible labels.
- Keyboard navigation must work throughout the site.
- Maintain visible focus states.
- Maintain sufficient color contrast.
- Provide alt text for meaningful images.
- Use ARIA only when native HTML semantics are insufficient.
- Do not communicate information through color alone.

28. Frontend Form States

- Default.
- Focused.
- Invalid.

- Submitting.
- Success.
- Server error.
- Network error.
- Disabled.
- Forms should prevent duplicate submissions while a request is in progress.

29. SEO Frontend Requirements

- Unique title and meta description per page.
- Semantic heading hierarchy.
- Canonical URLs.
- Open Graph and social metadata.
- Structured data where appropriate.
- Descriptive image alt text.
- Internal linking between services, solutions, portfolio and blog.
- Fast page rendering and Core Web Vitals optimization.

30. Performance Requirements

- Optimize images and use responsive image sizes.
- Lazy-load below-the-fold media.
- Minimize client-side JavaScript.
- Prefer server components/static generation for appropriate content.
- Use caching for public content.
- Avoid unnecessary third-party scripts.
- Target strong Lighthouse and Core Web Vitals performance.

31. Frontend Security

- Never place private API keys in client-side JavaScript.
- Send sensitive operations through secure server endpoints.
- Validate data server-side even when frontend validation exists.
- Do not render unsanitized HTML from user input.
- Use secure authentication flows for admin pages.
- Use Content Security Policy and other headers where practical.

32. Frontend State & Data Handling

- Use server state for public CMS data where appropriate.
- Use local component state for UI-only interactions.
- Persist only necessary client-side data.

- Do not store sensitive credentials in localStorage.
- Show skeletons/loading states for asynchronous content.
- Handle failed API requests gracefully.

33. Browser Support

- Support current versions of Chrome, Edge, Firefox and Safari.
- Test Chromium-based browsers on Windows and macOS.
- Verify mobile Safari and Chrome on common mobile devices.
- Gracefully degrade non-essential visual effects on unsupported environments.

34. Frontend Folder Structure

- app/ — routes and page layouts.
- components/ — reusable components.
- components/ui/ — shared UI primitives.
- components/sections/ — homepage and page sections.
- lib/ — utilities, API clients and validation.
- types/ — TypeScript types.
- public/ — static assets.
- styles/ — global styles/design tokens if needed.
- tests/ — frontend unit and end-to-end tests.

35. Frontend Acceptance Criteria

- All public pages render correctly on mobile, tablet and desktop.
- Navigation works and active states are visible.
- Every primary CTA leads to a functional destination.
- Contact, consultation and estimator flows have complete validation and success/error states.
- Desktop Application Development is represented consistently across navigation, service cards, service page, estimator and relevant landing pages.
- Images are optimized and do not create layout shifts where practical.
- Accessibility checks show no critical keyboard or labeling issues.
- SEO metadata exists for all intended indexable pages.
- No private API key is present in the browser bundle.
- Frontend build passes TypeScript checks, linting and automated tests.

36. Recommended Frontend Delivery Phases

- Phase 1: Design system, navigation, homepage, services, desktop app service, portfolio, about, process and footer.
- Phase 2: Contact, consultation, estimator, SEO, responsive polish and analytics.
- Phase 3: Blog, CMS-connected content, admin integration and advanced animations.

- Phase 4: Conversion optimization, A/B testing infrastructure where appropriate and additional landing pages.

Frontend Definition of Done

The CodeNova frontend is ready for production when the complete public website is responsive, accessible, SEO-ready, secure, performant, connected to the required backend flows and tested across supported browsers and screen sizes.

This document complements the CodeNova PRD, TRD and Security & Privacy Document. The implementation should follow the actual approved brand assets, business information and final content rather than placeholder claims.